

Cell Design Specification - GridStore-D3

VBF-T2R1-DS-01 | t2_r1 grid energy storage | 2026-08-25 | values mechanically sourced from parameter set / simulation outputs

1. Basic specification

Item	Value	Source
Electrochemical system	NMC811 / graphite, EC/EMC + LiPF6 (baseline chemistry; electrolyte transport overridden as formulation proxy)	Chen2020 parameter set
Nominal capacity	5.0 Ah (parameter set); 5.066 Ah simulated 1C	cell/r6_d3_1c.json:capacity_ah
Voltage window	2.5 - 4.2 V	Chen2020 dump
Discharge plateau proxy	3.722 V at fixed t = 1800 s (calc-energy 'midpoint' scalar is a step-density artifact)	cell/r6_d3_1c.json:voltage_v (fixed-time)
Cell dimensions	1580 x 65 x 0.2008 mm (W x H x T, layer stack)	Chen2020 dump; r6_d3_energy.json:thickness_m
Cell mass	39.63 g (contract caliber; electrolyte excluded)	cell/r6_d3_energy.json:mass_kg
Shell / casing / tabs	Not provided (no parameters in set)	Chen2020 dump
Electrolyte formulation	Baseline EC/EMC + LiPF6; transport as constants sigma 3.0 S/m, D 2.5e-9 m2/s, t+ 0.35 (formulation proxy; no molecular funnel - case started at stage 3)	candidates/r6_d3_params.json
Cation transference number	0.35 (base 0.2594)	r6_d3_params.json; Chen2020 dump

2. Electrode and separator

Layer	Thickness	Porosity (final)	Material / notes	Source
Positive electrode	75.6 um	0.40 (base 0.335)	NMC811, active vf 0.665, r_p 2.5 um (base 5.22 um), rho 3262	Chen2020 dump; r6_d3_params.json
Negative electrode	85.2 um	0.40 (base 0.25)	graphite, active vf 0.75, r_n 2.0 um (base 5.86 um), rho 1657	Chen2020 dump; r6_d3_params.json
Separator	12 um	0.47	polyolefin, rho 397	Chen2020 dump
Positive current collector	16 um	-	Al, rho 2700	Chen2020 dump
Negative current collector	12 um	-	Cu, rho 8960	Chen2020 dump
N/P ratio	= 1.98	-	(33133x0.9014x85.2um)/(63104x0.2700x75.6 um), initial-Li-content caliber; baseline geometry unchanged	Chen2020 dump, mechanical

3. Process design parameters

Parameter	Value	Formula / note
Positive areal density	147.96 g/m2	75.6um x 0.60 x 3262
Negative areal density	84.71 g/m2	85.2um x 0.60 x 1657
Positive compaction density	1.957 g/cm3	3262 x 0.60 / 1000

Parameter	Value	Formula / note
Negative compaction density	0.994 g/cm3	1657 x 0.60 / 1000
Electrolyte fill amount	8.62 g	pore volume 7.185 cm3 x 1.2 g/cm3 (literature value) x 100% fill
Formation recommendation	0.1C CC to 4.2V, 25C, 2 cycles	design-recommended; production value requires tuning

4. Mass breakdown (contract caliber, electrolyte excluded)

Layer	kg/m2	g/cell	Source
Positive electrode	0.147964	15.196	r6_d3_energy.json:layer_kg_m2 x 0.1027 m2
Negative electrode	0.084706	8.699	idem
Positive current collector	0.0432	4.437	idem
Negative current collector	0.10752	11.042	idem
Separator	0.002525	0.259	idem
TOTAL	0.385915	39.63 g	r6_d3_energy.json:mass_kg

5. Performance verification (vs entry-0 criteria)

Item	Value	Criterion	Determination	Source
Energy density	465.62 Wh/kg	>= 327.18 Wh/kg	PASS	cell/r6_d3_energy.json:energy_density_wh_kg
4C fast charge - plating	anode pot min +0.0498 V	no plating (>= 0 V)	PASS	cell/r6_d3_4c.json:anode_potential_v
SEI @100 cyc	9.087 nm	<= 500 nm	PASS	cell/r6_d3_aging100.json:sei_thickness_nm_end
SEI @500 cyc	25.317 nm	<= 550 nm	PASS	cell/r6_d3_aging500.json:sei_thickness_nm_end
-20C retention	99.571%	>= 90%	PASS	cell/r6_d3_lowT_ret.json (100 x 5.04452/5.06626 Ah)
-20C true-soak cross-check	99.571%, T_max 266.7 K	>= 90%	PASS	cell/r6_d3_lowT_soak.json (Initial temperature 253.15 K probe)
4C acceptance / temperature	0.838 Ah; T_max 342.4 K (+24.2 K)	informational	-	cell/r6_d3_4c.json

6. Design notes

1) Electrolyte transport constants (σ 0.4->3.0 S/m; D 4e-10->2.5e-9 m2/s; t_+ 0.2594->0.35): baseline 4C plated (min -0.1918 V); constant formulation proxy solved plating (R2) and enables -20C. 2) Negative particle radius 5.86->2.0 μ m (R2/R4): 4C margin +0.091 V, acceptance 0.553 Ah. 3) Positive particle radius 5.22->2.5 μ m (R5): +50% acceptance (0.828 Ah), ED 458.1 Wh/kg. 4) k_{sei} 1e-12->1e-16 m/s (R3, coating bridge): SEI@500cyc 777.9->21.7 nm. 5) Negative porosity 0.35->0.40 (R6 probe): strictly non-worse; ED 465.6 Wh/kg; margin +0.0498 V (within solve noise).

Honest annotations: volume-fraction consistency (active vf 0.665/0.75 defined at base porosity; final porosity 0.40 makes active+pore>1 in model bookkeeping - interpreted as replacing unmodeled binder/additive volume); calc-energy midpoint/DCR scalars are solver-step-density index artifacts (time-integral metrics unaffected; fixed-time v@1800s used); aging per-cycle capacity is a PyBaMM variable artifact (probe: real steps are full 1C swings 5.08/4.75 Ah; sei_thickness_nm_end accumulates over genuine full-depth cycles); part of the porosity ED gain is contract-caliber (electrolyte excluded from mass; worst-case re-inclusion still 459.6 Wh/kg); 4C temperature +24.2 K above ambient with no contract red line (cooling h not a design lever).